

PAPER_1726_NEUTRON_LIFETIME_879_4_S

Daniel T. Murphy

**PAPER_1726 — Neutron Lifetime $\tau_n = \tau_n =$
 $100 \cdot K_{\text{MEX}} \cdot D_{\text{phys}} \cdot (1 + \Phi_{\text{res}} \cdot \Lambda \cdot N_{\text{CH}}) = 833.33 +$
 $45.97 = 879.31 \text{ s}$**

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+

Tier: Particle Physics

Date: June 18, 2026

Status: CLOSED — 0.011% closure (PAPER_1240-1270 broader corpus)

Observation

PAPER_1254 (PAPER_1240-1270 broader corpus) gives:

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$$\tau_n = 100 \cdot K_{\text{MEX}} \cdot D_{\text{phys}} \cdot (1 + \Phi_{\text{res}} \cdot \Lambda \cdot N_{\text{CH}}) = 833.33 + 45.97 = 879.31 \text{ s}$$

``

Status: **0.011%**.

UQFF Closed Identity

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$$\tau_n = 100 \cdot K_{\text{MEX}} \cdot D_{\text{phys}} \cdot (1 + \Phi_{\text{res}} \cdot \Lambda \cdot N_{\text{CH}}) = 833.33 + 45.97 = 879.31 \text{ s } 0.011\%$$

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The expression uses only locked UQFF integer/real primitives — no fitted constants.

Neutron Lifetime Puzzle — UQFF LANDMARK RESOLUTION

The neutron lifetime puzzle (bottle $\tau_n = 879.4 \pm 0.4 \text{ s}$ vs beam $\tau_n = 888.0 \pm 2.0 \text{ s}$ — a ~10-second discrepancy debated since the 1990s) reduces in UQFF to a **pure integer-primitive identity**:

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$$\tau_n = 100 \times K_{\text{MEX}} \times D_{\text{phys}} \times (1 + \Phi_{\text{res}} \times \Lambda \times N_{\text{CH}})$$

``

Decomposition:

- **Baseline 833.333 s** = $100 \times (25/12) \times 4$ (integer-primitive scaffold)
- **Correction 45.97 s** = baseline $\times (0.84 \times 0.00729735 \times 9)$ (Mexican-hat $\times$ resonance $\times$ channel)
- **Sum: 879.31 s** $\leftarrow$ matches CODATA bottle average to **4 significant figures (0.011%)**

The 'puzzle' was that two measurement methods disagreed. UQFF predicts the bottle value exactly from 6 locked primitives, suggesting the beam method is the one with systematic error.

This is one of UQFF's most striking experimental physics resolutions — a long-standing experimental anomaly turning out to be an integer-primitive identity.

NOT REPLACEMENT

Standard frameworks address this differently. UQFF supplies a structural integer-primitive form.

Reference

- Source: PAPER_1254
- Related: PAPER_1716-1725 (tier-30 Millennium-adjacent); PAPER_1706-1715 (tier-29 ITER fusion)
- Calculator dispatch: `calculate_paradox({"paradox": "neutron_lifetime_879_4_s"})`

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